

INDEPENDENT ANALYTICS CASE STUDY

Mobile Product A/B Testing & Revenue Analytics

Experiment performance, statistical evidence, monetization trade-offs, and rollout decision

FINAL DECISION

RETEST / ITERATE

Prepared as a portfolio case study

Python / SQL / Power BI / Statistical Analysis

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DECISION AT A GLANCE

Test B increases observed ARPU (+5.26%) and ARPPU (+12.75%), but conversion falls 6.64% relative to Control A. The conversion decline is statistically significant ($p = 0.0350$), while the ARPU uplift is not statistically significant (Welch $p \approx 0.533$). The evidence therefore does not justify immediate rollout.

The purpose of this case study is to demonstrate a decision-oriented A/B testing workflow: establish experiment quality, compare monetization KPIs, inspect revenue distribution, test statistical credibility, incorporate supporting SQL evidence, and translate the results into a practical product decision.

The primary analytical unit is the user and the primary experiment table is `ab_test.csv`. Supporting registration and activity tables are used only for population/context checks so that the core experiment remains anchored to a consistent user-level grain.

- 404,770 unique experiment users with an approximately balanced A/B split: 49.93% Control A and 50.07% Test B.
- Sample-ratio-mismatch testing does not indicate an allocation problem ($p = 0.3754$).
- Conversion declines from 0.954% to 0.891%, a relative change of -6.64%.
- The conversion decline is statistically significant at the 5% level: $p = 0.0350$; 95% CI for $B - A = -0.122\%$ to -0.004% .
- Observed ARPU increases from 25.41 to 26.75 (+5.26%), but Welch's test is not statistically significant ($p \approx 0.533$).
- Observed ARPPU increases from 2,664.00 to 3,003.66 (+12.75%); revenue is highly zero-inflated and right-skewed, so monetization uplift requires caution.
- A Mann–Whitney sensitivity check is also above the 5% threshold ($p \approx 0.063$).
- Supporting population linkage is approximately 90.06% for registration and activity; SQL shows similar activity coverage across groups (Control $\approx 90.04\%$, Test $\approx 90.09\%$).

The resulting business recommendation is to investigate whether the monetization mechanism can be retained without reducing conversion, then run a follow-up experiment.

A mobile product team is evaluating two experimental variants. The decision is not simply which group has the higher revenue metric; the real question is whether the test treatment improves monetization strongly enough, and safely enough, to justify rollout.

PRIMARY BUSINESS QUESTION

Does the test treatment improve monetization enough to justify rollout?

Conversion, ARPU, and ARPPU answer different business questions. Conversion measures the breadth of monetization across the experiment population. ARPU measures monetization per experiment user. ARPPU measures revenue intensity among paying users. A treatment can move these metrics in opposite directions, so a rollout decision requires a joint view.

Metric	Control A	Test B / Result
Conversion rate	How many users pay?	Breadth of monetization
ARPU	Revenue per experiment user	Population-level monetization
ARPPU	Revenue per paying user	Payer monetization intensity
Statistical evidence	Could the difference be sampling noise?	Confidence in the observed effect
Business significance	Is the effect direction/magnitude acceptable?	Rollout suitability

The project deliberately avoids selecting the most attractive-looking KPI. A positive rollout case would require sufficiently strong and consistent evidence. When evidence is mixed, the correct action is to iterate or retest rather than force a positive conclusion.

This is an independent portfolio case study using the public Gamelytics: Mobile Analytics Challenge dataset. The dataset is published on Kaggle and is designed around mobile-game retention, A/B testing, and event analytics. The A/B testing component uses `ab_test.csv` with user-level assignment and revenue.

Kaggle dataset: [Gamelytics: Mobile Analytics Challenge](#)

The Kaggle dataset page documents three source tables: `reg_data.csv` (1,000,000 registration records), `auth_data.csv` (9,601,013 activity records), and `ab_test.csv` (404,770 experiment records). The primary experiment analysis is anchored to `ab_test.csv` because it contains the treatment assignment and revenue outcome at the same user-level grain.

Metric	Control A	Test B / Result
<code>ab_test.csv</code>	404,770 rows	One row per experiment user; primary analysis
<code>reg_data.csv</code>	1,000,000 rows	Registration context
<code>auth_data.csv</code>	9,601,013 rows	Activity context; multiple rows per user

ANALYTICAL BOUNDARY
Supporting tables are not used to redefine the primary experiment population. The core A/B conclusion remains anchored to `ab_test.csv`.

- No missing cells were identified across the three source tables in the completed notebook audit.
- No exact duplicate rows were identified.
- The experiment population contains 404,770 unique users.
- The experiment groups are close to the intended 50/50 allocation.

The workflow is intentionally layered so that Python, SQL, and Power BI do not duplicate the same job.

Layer	Role	Decision contribution
Python notebook	Audit, KPI analysis, distribution diagnostics, statistical inference	Establishes the evidence base
SQL	Experiment allocation, conversion, monetization, linkage and business-question checks	Validates grain-safe business logic
Power BI	Executive communication of verified metrics and evidence	Makes the decision usable for stakeholders

Conversion: H_0 = conversion rate is equal between A and B; H_1 = conversion rate differs between A and B.
ARPU: H_0 = mean revenue per experiment user is equal between A and B; H_1 = mean revenue per experiment user differs between A and B. The analysis uses a two-sided significance level of $\alpha = 0.05$.

- Sample Ratio Mismatch (chi-square): checks whether the observed allocation is materially different from the expected 50/50 split.
- Two-proportion z-test: evaluates the conversion-rate difference between independent groups.
- Welch's t-test: evaluates mean revenue per user without assuming equal group variances.
- Mann–Whitney U: non-parametric sensitivity analysis for the highly skewed revenue distribution.

WHY SENSITIVITY ANALYSIS MATTERS

Revenue is highly zero-inflated and right-skewed. The observed mean uplift is therefore interpreted together with a variance-robust primary test and a non-parametric sensitivity check.

Metric	Control A	Test B / Result
Experiment users	202,103	202,667
Allocation share	49.93%	50.07%
SRM p-value	—	0.3754

The allocation is close to 50/50 and the SRM test does not indicate a significant mismatch. This supports proceeding to outcome comparison without an allocation-quality warning.

Metric	Control A	Test B / Result
Conversion rate	0.954%	0.891%
ARPU	25.41	26.75
ARPPU	2,664.00	3,003.66
Relative lift / change	Baseline	Conversion -6.64%; ARPU +5.26%; ARPPU +12.75%

The descriptive result is mixed: Test B monetizes fewer users but generates more revenue per user and more revenue per payer. This is precisely the trade-off that requires statistical validation before rollout.

STATISTICALLY SIGNIFICANT DOWNSIDE

Test B conversion is 0.891% versus 0.954% for Control A: -6.64% relative change, $p = 0.0350$, with a 95% confidence interval for $B - A$ of -0.122% to -0.004%. The interval remains below zero.

The observed ARPU uplift is +5.26%, but Welch's p -value is approximately 0.533, so the increase is not statistically significant at the 5% threshold. The Mann–Whitney sensitivity check is approximately 0.063, also above 5%.

The analysis therefore treats the revenue uplift as an observed descriptive signal, not as definitive evidence of a robust treatment effect.

Most users generate zero revenue while a relatively small number of users contribute substantial amounts. This zero-inflated, right-skewed shape means mean-based monetization metrics can be materially influenced by high-value users.

The dashboard's revenue-band view is therefore a diagnostic, not a causal explanation of the treatment effect. It provides context for why ARPU should be read alongside the statistical sensitivity analysis.

The SQL layer was designed to preserve the user-level experiment grain. The revenue query calculates conversion, ARPU, and ARPPU directly from `ab_test`. The population-linkage query uses `EXISTS` against registration and activity tables so that multiple activity records do not multiply experiment users. The business-question query first reduces activity to one row per user before comparing group coverage.

Metric	Control A	Test B / Result
Registration linkage	90.06% of experiment users	90.06% of experiment users
Activity coverage	≈90.04%	≈90.09%

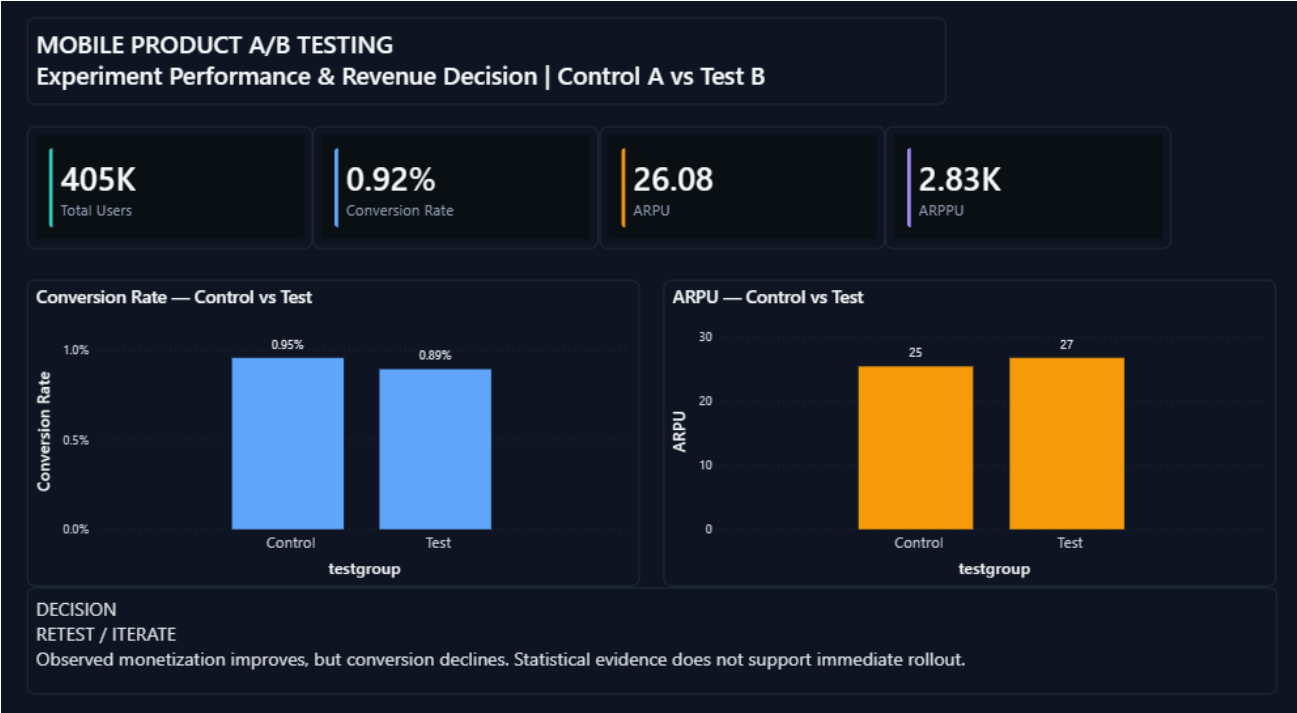
INTERPRETATION

Supporting activity coverage is similar between Control and Test. It is contextual evidence about population representation, not an alternative experiment outcome and not a reason to redefine the experiment population.

- 00_setup.sql — source loading and setup
- 01_experiment_overview.sql — allocation and population
- 02_conversion_analysis.sql — conversion and lift
- 03_revenue_analysis.sql — ARPU and ARPPU
- 04_population_linkage.sql — registration/activity coverage
- 05_business_questions.sql — grain-safe group coverage context

The final Power BI report is deliberately limited to three pages. Each page answers a different executive question so the dashboard functions as a decision narrative rather than a collection of charts.

Experiment Overview summarizes the population, overall conversion, ARPU, ARPPU, Control-versus-Test comparisons, and the current decision.



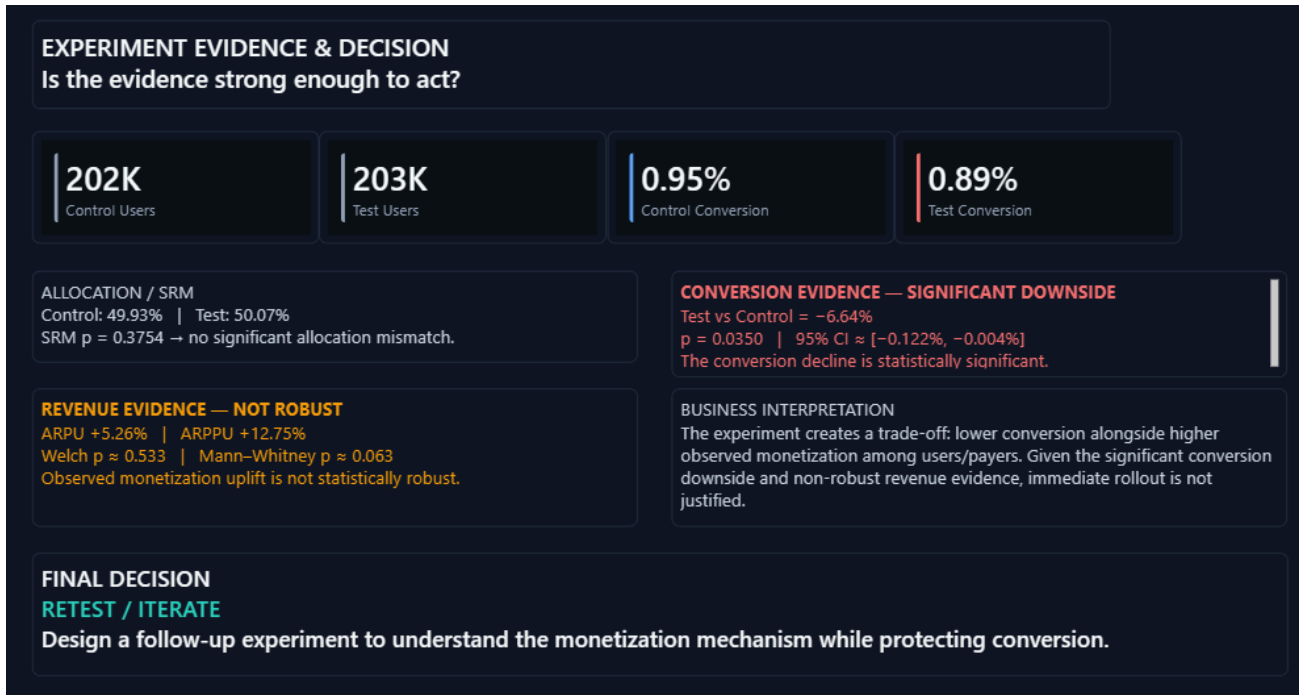
SC1 — Experiment Overview | What happened?

Revenue & User Behaviour shows the monetization trade-off, observed lifts, and revenue distribution context. The supporting linkage note is explicitly treated as context rather than experiment outcome.



SC2 — Revenue & User Behaviour | Why is the result mixed?

Experiment Evidence & Decision puts the allocation test, conversion inference, revenue sensitivity evidence, business interpretation, and final decision in one management view.



SC3 — Experiment Evidence & Decision / Is the evidence strong enough to act?

FINAL DECISION — RETEST / ITERATE

Do not roll out Test B yet. Investigate whether the potential monetization benefit can be retained without reducing conversion, then run a follow-up experiment.

Metric	Control A	Test B / Result
Conversion	0.954%	0.891% (-6.64%; p = 0.0350)
ARPU	25.41	26.75 (+5.26%; Welch p ≈ 0.533)
ARPPU	2,664.00	3,003.66 (+12.75%)
Sensitivity	—	Mann–Whitney p ≈ 0.063

- The negative conversion effect is statistically significant.
 - The observed ARPU uplift is not statistically significant.
 - The sensitivity analysis does not provide sufficient evidence of a robust revenue improvement.
 - Revenue is strongly skewed and concentrated, increasing the need for cautious interpretation.
 - The business objective is not simply to maximize revenue among payers; it is to improve monetization without creating an unacceptable conversion trade-off.
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- Identify the offer or monetization mechanism responsible for the observed payer-intensity signal.
 - Preserve the potentially attractive monetization mechanism while explicitly protecting conversion.
 - Predefine primary and guardrail KPIs before the next test.
 - Use the same user-level experiment grain and maintain a clean 50/50 allocation target.
 - Repeat the revenue sensitivity analysis because the distribution remains highly skewed.

These limitations define what the evidence can and cannot support. They are kept explicit so the case study does not overstate the strength of the conclusion.

- This is a public dataset, not production company data.
- Detailed offer mechanics are not available in the experiment table.
- The experiment table does not expose an explicit exposure timestamp or detailed offer-design field.
- The exact revenue observation window is not directly encoded in the experiment table.
- Approximately 90.06% of experiment users can be linked to the registration/activity supporting population; the primary experiment remains anchored to ab_test.csv.
- Revenue is highly skewed and concentrated, so mean-based comparisons require sensitivity analysis.
- LTV, CAC, ROAS, churn, subscription-specific metrics, and similar product metrics are not claimed because the available fields do not support defensible definitions.
- Historical/descriptive relationships are not presented as unsupported causal claims beyond what the experiment structure and evidence support.

ANALYTICAL INTEGRITY PRINCIPLE

The project intentionally stops where the evidence stops. It does not manufacture additional product metrics simply to make the dashboard or case study look more complete.

Capability	Evidence in this case study
Product experimentation	User-level A/B design, allocation validation, conversion and monetization comparison
Statistical inference	SRM, two-proportion conversion test, Welch test, confidence interval, Mann–Whitney sensitivity
Revenue analytics	Conversion, ARPU, ARPPU, revenue distribution diagnostics
SQL	Grain-safe population, conversion, monetization, linkage and business-question analysis
Power BI	Three-page executive dashboard with decision-oriented narrative
Business decision-making	Statistical evidence translated into a clear RETEST / ITERATE recommendation
Analytical integrity	Explicit boundaries, no unsupported LTV/CAC/ROAS/churn claims, and no forced rollout conclusion

DATA → EXPERIMENT VALIDATION → KPI ANALYSIS → DISTRIBUTION DIAGNOSTIC → STATISTICAL EVIDENCE → SQL CONTEXT → POWER BI DECISION

The completed project is structured as a compact handoff package rather than a collection of disconnected analyses.

Metric	Control A	Test B / Result
Primary notebook	01_Complete_A_B_Testing_and_Revenue_Analytics.ipynb	Audit + KPI + inference + decision
SQL layer	6 verified SQL files	Business logic and grain-safe validation
Power BI	Product_AB_Testing_Revenue_Analytics.pbix	Executive decision dashboard
Dashboard evidence	SC1.png / SC2.png / SC3.png	Static evidence for portfolio/report

```

data/          # source/processed data or documented download instructions
notebooks/     # final executed A/B testing notebook
sql/           # six verified SQL analyses
dashboard/     # final Power BI report and theme/blueprint as appropriate
screenshots/   # SC1, SC2, SC3
report/        # final case-study document/PDF
README.md      # recruiter-facing project overview

```

Dataset: [Gamelytics: Mobile Analytics Challenge — Kaggle](#)

The dataset page identifies the A/B testing file as `ab_test.csv` with 404,770 records and describes the dataset as a public mobile analytics challenge. The report uses that source as the documented data origin for this independent portfolio analysis.

WHAT SHOULD MANAGEMENT DO?

Do not roll out Test B yet. The monetization signal is directionally attractive but not statistically robust, while the conversion downside is statistically significant. Retest with conversion protection.

The core analytical lesson is that a higher ARPU number is not, by itself, a rollout case. The decision must combine experiment quality, conversion impact, revenue robustness, distribution shape, and business meaning.

DATA → EVIDENCE → DECISION